Kron's Schrödinger network

The electron of a chemical bond (H_2^+) solved as an electric network after Kron 1945, with the H atom as its check. CLAUDE.md §3a.1

kronspice simulation report · 2026-11-07 · atomic units; grids h = 0.2, 0.1, 0.05 a0

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Summary

Kron (*Phys. Rev.* 67, 39, 1945) wrote the Schrödinger equation as an electric network: series inductors carry the kinetic energy, shunt inductors to ground the potential energy, and shunt capacitors the energy eigenvalue. The wave function ψ is the junction voltage to ground, and an eigenstate is a resonance of the network. This example builds that network for H_2^+, the simplest chemical bond (two protons, one electron), with the hydrogen atom as its check; both have exact solutions. It is the electronic side of the chemistry plan (the vibrational side is *water_vibration*).

- **Hydrogen atom:** E → −0.499728 Ha = −13.5983 eV as h → 0 (order 1.44), against the exact −0.5 Ha = −13.6057 eV.
- **H_2^+ at R = 2 a_0:** electronic energy E → −1.102291 Ha (order 1.46), against the exact −1.1026342 Ha.
- **The bond:** length R_e = 1.998 a_0 = 1.057 Å (exact 1.997 a_0 = 1.057 Å) and energy D_e = 0.1024 Ha = 2.786 eV (exact 0.1026 Ha = 2.79 eV).
- **Circuit twin:** the same network as an LC circuit in kronspice resonates at E = −1.13848455 Ha on the coarse grid (h = 0.4 a_0, 1950 junctions), equal to the driven-network eigenvalue −1.13848455 Ha (test tolerance 10^{-7}).
- **Torn solve equals MNA:** the 1950-node circuit torn into 4 pieces joined by 179 links matches the full solve to 3.1×10^{-15} of the largest node voltage, with the same power 0.302832 at phase −90.00°.

The construction

The ground state is axially symmetric (m = 0), so the network is laid out on the (ρ, z) half-plane with the nuclei on the axis. In atomic units, the equation multiplied by ρ so the network is reciprocal is

$$-\tfrac{1}{2}\,[\partial_\rho(\rho\,\partial_\rho\psi) + \rho\,\partial_z^2\psi] + \rho\,V\,\psi = E\,\rho\,\psi, \quad V = -\Sigma\,Z_a\,/\,|r - R_a|$$

where Z_a and R_a are the nuclear charges and positions. On a grid of spacing h with junctions at $\rho_i = (i + \tfrac{1}{2})h$ (no junction on the axis, so it needs no boundary condition) this is A ψ = E B ψ with B = diag(ρ_i). In Kron's network: series branches of admittance $\rho_{face}/(2h^2)$ between neighbours, a shunt branch $\rho_i \cdot V_i$ at each junction and a shunt capacitor ρ_i. The outer boundary is a short circuit (ψ = 0) far from the nuclei, Kron's termination "by a short circuit at a point of zero voltage".

The eigenvalue is found two ways:

- **Kron's driven network.** The network is driven at a trial energy σ by sources B·x and its response is the next x (inverse iteration with the factors computed once); the response converges to the resonance nearest σ and the Rayleigh quotient gives E. Energies at h, h/2 and h/4 are extrapolated to h → 0 (Richardson, with the order estimated from the three runs).
- **A kronspice LC circuit.** The same network written as a SPICE netlist in its dual form (capacitors and inductors exchanged, because Kron's all-inductor network has loops of shorts at DC): series capacitors between junctions, shunt capacitors ρ(V + V0) and shunt inductors 1/ρ, with V0 shifting the potential so every element is positive. Every eigenstate is then a resonance at ω² = 1/(E + V0), the ground state the highest, and kronspice's AC analysis finds it.

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Results

Finite-difference energies lie below the exact ones (a difference scheme is not variational; the grid samples the Coulomb wells too deeply) and converge with an order set by the cusps, about 1.45 on these grids.

Hydrogen atom (check)

h (a_0)	junctions	E (Ha)
0.200	9,800	−0.51087970
0.100	39,200	−0.50382944
0.050	156,800	−0.50123643
h → 0 (order 1.44)		**−0.499728 = −13.5983 eV (exact −0.5 Ha = −13.6057 eV)**

H_2^+ at R = 2 a_0, electronic energy

h (a_0)	junctions	E (Ha)
0.200	7,800	−1.11745397
0.100	31,200	−1.10780517
0.050	124,800	−1.10429632
h → 0 (order 1.46)		**−1.102291 (exact −1.1026342)**

The bond curve

Total energy E(R) + 1/R (the 1/R term is the protons' repulsion), each R extrapolated to h → 0 from h = 0.2 and 0.1, with a least-squares cubic through the minimum:

R (a_0)	R (Å)	total energy (Ha)
1.6	0.847	−0.590490
1.8	0.953	−0.599866
2.0	1.058	−0.602291
2.2	1.164	−0.600531
2.4	1.270	−0.596271
2.8	1.482	−0.584110

	model	exact
bond length R_e	1.998 a_0 = 1.057 Å	1.997 a_0 = 1.057 Å
bond energy D_e	0.1024 Ha = 2.786 eV	0.1026 Ha = 2.79 eV

Circuit twin and torn solve

- Coarse grid, h = 0.4 a_0, 1950 junctions: kronspice AC resonance E = −1.13848455 Ha; driven-network eigenvalue on the same grid −1.13848455 Ha. The test requires agreement to 10^{-7}.
- The exported netlist (*kron_schrodinger_h2plus_coarse.cir*, 7831 elements, 1950 nodes) is solved at AC (0.078178 Hz) by automatic tearing into 4 pieces of 976, 974, 976 and 974 unknowns, joined by 179 links; the only coupled solve is of order 179, against 3900 unknowns for the whole circuit.
- Torn against full (MNA) solve: largest node-voltage difference 3.1×10^{-15} of the largest node voltage; power taken by the R, L, C branches 0.302832 at phase −90.00° both torn and untorn (power is invariant under tearing, Roth 1959 §5).

Validation

- *main_test.go*: convergence to the exact energies; circuit resonance equals the eigenvalue; cubic minimum.
- Exact references: Bates, Ledsham and Stewart (1953) and Peek (1965) for H_2^+; CODATA 2018 for the hartree and the bohr.

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Limits

- Only the axially symmetric (m = 0) ground state is treated, on a 2-D (ρ, z) grid.
- Finite-difference energies are not variational and lie below the exact ones; the h → 0 values rely on Richardson extrapolation with an order (about 1.45) set by the Coulomb cusps.
- One electron only. Water has ten interacting electrons: the next step is a self-consistent field (Hartree–Fock) on a 3-D network, with a Poisson network for the repulsion, which is not done here.

Reproducing the results

```
go run ./examples/kron_schrodinger
go test ./examples/kron_schrodinger
```

Sources: G. Kron, "Electric circuit models of the Schrödinger equation," Phys. Rev. 67, 39–43 (1945); D. R. Bates, K. Ledsham, A. L. Stewart, Phil. Trans. R. Soc. A 246, 215 (1953); J. M. Peek, J. Chem. Phys. 43, 3004 (1965); CODATA 2018.

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